Postman Notes

Introduction to Postman

Postman is a popular API testing and development tool used by developers to design, test, debug, and document APIs easily.

It provides a graphical user interface (GUI) to send HTTP requests and receive responses without writing complex code.

Features of Postman

1. API Testing

- Send requests to APIs
- Check API responses
- Validate status codes and data

2. User-Friendly Interface

- Easy GUI
- No need to use command line tools

3. Supports Multiple HTTP Methods

- GET
- POST
- PUT
- DELETE
- PATCH

4. Collections

- Save API requests in groups
- Reuse requests easily

5. Environment Variables

- Store values like:
    - URLs

- Tokens
- Usernames
- Passwords

6. Automation Testing

- Write test scripts using JavaScript
- Automate API validation

7. API Documentation

- Generate and share API documentation

8. Collaboration

- Team members can share collections and environments

Installation of Postman

1. Visit the official website:

[Postman Official Website]

2. Download the application
3. Install it on:
   - Windows
   - Linux
   - macOS

HTTP Methods in Postman

Method	Purpose
GET	Retrieve data
POST	Send new data
PUT	Update existing data

Method	Purpose
DELETE	Remove data
PATCH	Partially update data

Components of Postman Interface

1. Request Section

Used to:

- Enter API URL
- Select HTTP method
- Add headers and body

2. Response Section

Displays:

- Status code
- Response body
- Headers
- Response time

3. Collections

Stores saved requests.

4. Environment

Stores variables for different systems.

Creating a GET Request

Steps

1. Open Postman
2. Select **GET**

3. Enter API URL
4. Click **Send**
5. View the response

Example API

https://jsonplaceholder.typicode.com/posts

Creating a POST Request

Steps

1. Select **POST**
2. Enter API URL
3. Go to **Body**
4. Select **raw → JSON**
5. Enter JSON data
6. Click **Send**

Example JSON

```
{
  "title": "Test Post",
  "body": "Hello World",
  "userId": 1
}
```

Status Codes

Code	**Meaning**
200	OK
201	Created

Code	Meaning
400	Bad Request
401	Unauthorized
404	Not Found
500	Internal Server Error

Headers in Postman

Headers provide additional information about the request.

Example

Content-Type: application/json

Authorization: Bearer token

Authorization Types

Type	Description
Basic Auth	Username and password
Bearer Token	Token-based authentication
API Key	Unique access key
OAuth 2.0	Secure authentication framework

Environment Variables

Environment variables help avoid repeating values.

Example

Variable	Value
base_url	https://api.example.com

Variable Value

token abc123

Usage:

{{base_url}}/users

Writing Tests in Postman

Postman supports JavaScript for testing.

Example Test Script

```
pm.test("Status code is 200", function () {
  pm.response.to.have.status(200);
});
```

Collections in Postman

Collections are used to:

- Organize requests
- Share APIs
- Run automated tests

Benefits:

- Easy maintenance
- Reusability
- Team collaboration

Advantages of Postman

- Easy to use
- Supports automation
- Fast API testing

- Good collaboration features
- Supports multiple environments
- Helps in debugging APIs

Limitations of Postman

- Large collections may become slow
- Requires internet for some cloud features
- Advanced automation may require coding knowledge

Common Uses of Postman

- API development
- API testing
- Backend debugging
- Automation testing
- API documentation
- Team collaboration

Conclusion

Postman is one of the most widely used API testing tools in software development. It simplifies API creation, testing, debugging, and documentation with an easy-to-use graphical interface. It is highly useful for developers, testers, and DevOps engineers.